



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY OF NEURAL CIRCUITS

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Abstract

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Chapter 1

Stochastic Resonance in the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1–3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical stochastic resonance

The idea of studying the effects of noise on the dynamics of a system first appeared parallely in the work of Benzi et al. [5] and Nicolis [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature.

A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = R_{in}(T) - R_{out}(T) \tag{1.1}$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$\begin{aligned} R_{in} &= Q\mu \\ R_{out} &= \alpha(T)Q\mu + \varepsilon(T) \end{aligned}$$

where μ is the average incoming solar radiation.

In [5] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mu\varepsilon(T)}{1 + \beta\,(1 - T/T_1)\,(1 - T/T_2)\,(1 - T/T_3)} - \varepsilon(T) \tag{1.2}$$

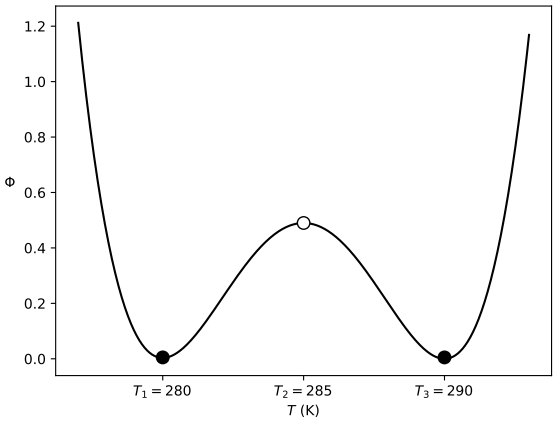


Figure 1.1: The double well potential. Filled circles are stable points.

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{\mathrm{d}T}{\mathrm{d}t} \ , \tag{1.3}$$

then a pseudo-potential energy function is defined as

$$\Phi(T) = -\int F(T)\mathrm{d}T \tag{1.4}$$

that represents the "energy" needed to change the temperature across temperature values. From Φ we get

$$F(T) = -\frac{\partial \Phi}{\partial T} \tag{1.5}$$

Then intuitively the maximum of Φ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{\mathrm{d}F}{\mathrm{d}T} = -\frac{\partial^2 \Phi}{\partial T^2} \tag{1.6}$$

is < 0, any neighboring starting state of the earth's temperature will converge to it, while when it is > 0 they will diverge.

It can be shown that T_1, T_3 are stable points, while T_2 is unstable: a peak any starting state would "roll off" from (see A). In our context, T_1, T_3 are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. A system where 2 stable states separated by an unstable state exist is called a bistable system, and the plot of Φ over T displays the typical double well shape of the (pseudo)potential function (Figure 1.1).

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., whether it is smaller or greater than T_2. Then it would follow $F(T)$ towards T_1 or T_3 (Figure 1.2). However, earth's temperatures do indeed show oscillations with a period of 10^5 years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then μ becomes time dependent

$$\mu(t) = 1 + \gamma \cos(\omega t) \tag{1.7}$$

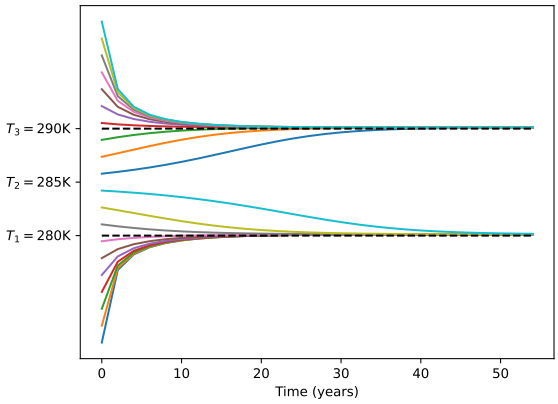


Figure 1.2: Trajectories for different starting temperatures without noise.

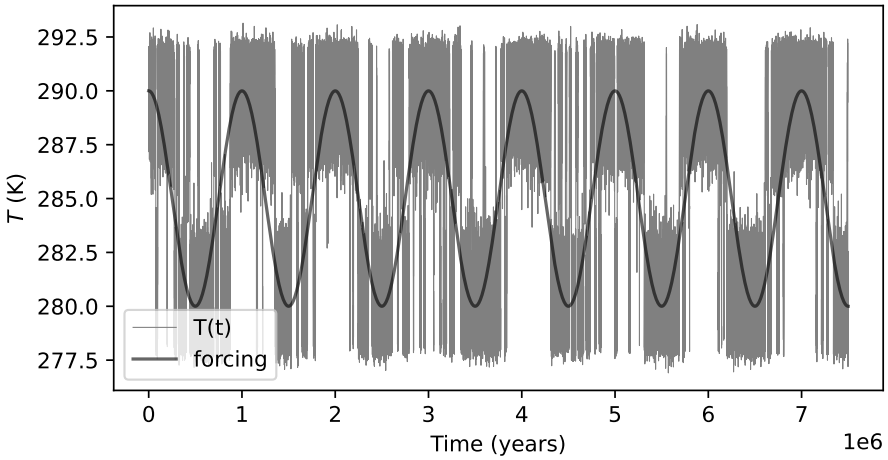


Figure 1.3: Temperature over time from numerical solution of 1.2, with $\sigma = 0.15$.

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become time dependent $\tilde{F} = \tilde{F}(T;t)$, $\tilde{\Phi} = \tilde{\Phi}(T;t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over T_2 (see fig).

If gaussian white noise, representing various mechanisms causing T fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \tilde{F}(T;t) + \sigma\eta(t) \tag{1.8}$$

then for certain values of σ a jump between stable states becomes possible, with a frequency $\approx 2\pi/10^5$. That is, the system shows temperature oscillations at the forcing frequency, as noise allows the system to jump from one equilibrium to another (Figure 1.3). Note that oscillations arise from the joint effect of the periodic forcing and noise: noise pushes the system away from the equilibrium at every timestep, but only when the forcing asymmetrically tilts the double well potential such that the distance between the current equilibrium and the unstable one becomes small enough to allow the stochastic component to push the system over it, a climate transition (that is, a jump from T_1 to T_3 or viceversa) can occur. This can be seen as an "escape" from a stable equilibrium, requiring a jump over a (pseudo)energy potential "wall". This is a classic problem in physics, Kramer's rate [7] gives the frequency of jumps per second

$$r_K = \frac{\sqrt{V''(x_m)/m}\sqrt{|V''(x_b)/m|}}{2\pi\gamma} e^{-\frac{\Delta V}{D}} \tag{1.9}$$

where $V''(x_m)$ is the curvature around the minima, $V''(x_b)$ that at the maximum, ΔV the distance from the minimum to the maximum, and $D = \sigma^2/2$ the temperature of the system (noise intensity in our context). The inverse $\frac{1}{r_K}$ gives the residence time, the time spent at one of the minima. If our condition for SR is that it allows the system to follow the subthreshold forcing, then the optimal D is such that, given the forcing period T_ω and the residence time as a function of D, $T_K(D) = \frac{1}{r_K(D)}$ is

$$T_K(D) = \frac{T_\omega}{2} \tag{1.10}$$

Some useful considerations can be made on SR defined on the climate model. First, it requires a system containing a cubic nonlinearity, as in 1.2, such that 2 stable states separated by a potential wall exist. Additionally, the periodic forcing must be insufficient alone to cause a crossing (subthreshold). Later it will be shown that both of these assumptions can be relaxed. Last, as 1.10 suggests, too low D will not be able to push the system over the barrier at the frequency needed, and high enough D make the potential barrier effectively nonexistent, increasing r_K and making the double well model meaningless.

Hence, given a metric of how well the system follows the forcing, there will be a peak for a certain D. In this sense, there is a resonance in the system. For classic SR models, the natural measure is the spectral power (obtained by Fourier transform of the system response) at the forcing frequency relative to the neighboring ones: Signal-to-Noise Ratio (SNR) [8].

1.1.2 Small detour on dynamical systems

It will be shown that neurons are better modeled by 2D systems, due to the coexistence and interaction of 2 main ionic currents that govern a neuron's state $\left(\mathrm{Na}^+ \text{ and } \mathrm{K}^+\right)$, that determine a depolarization-hyperpolarization feedback loop. To analyze a 2+D systems the tools of dynamical systems theory are indispensable. A brief covering of the main methods will follow, as they will be referenced in the following chapters (for a more complete tractation, see [9]).

For any 2D system defined by

$$\dot{v} = f(v, u) \tag{1.11}$$

$$\dot{u} = g(v, u) \tag{1.12}$$

the system exists in the space $S = \{v\mathbf{e}_1 + u\mathbf{e}_2 \mid v, u \in \mathbb{R}\}$, defined by a set of parameters $\mathcal{P}$, and always identified with the state vector $\mathbf{s} = (v, u)$.

Then we define the mapping $F : \mathbb{R}^2 \to \mathbb{R}^2$ such that $F : \mathbf{s} \mapsto (f(\mathbf{s}), g(\mathbf{s}))$, the standard definition of a vector field (in $\mathbb{R}^n$). The direction of $F(\mathbf{s})$ is given by the sign of $f(\mathbf{s})$ and $g(\mathbf{s})$. So the curves defined by $f(v, u) = 0$ and $g(v, u) = 0$, the v- and u-nullclines, that separate the plane in changes in the sign of $\dot{v}$ and $\dot{u}$, are of particular interest. Note that $\dot{v} = \mathrm{d}v(t)/\mathrm{d}t$ and $\dot{u} = \mathrm{d}u(t)/\mathrm{d}t$. Then the evolution of the system's state from an initial position $\mathbf{s}(t_0) = (v(t_0), u(t_0))$ up to a time t, $\mathbf{s}(t)$, is governed by the flow of the vector field, formally $\varphi^t : S \to S$. Indeed, the trajectory $\varphi^{t-t_0}(\mathbf{s}_0)$ is a curve always tangent to F.

All points of intersection of the nullclines are such that $\dot{v} = \dot{u} = 0$. Consider the state $\mathbf{s}^*$ such that $f(\mathbf{s}^*) = g(\mathbf{s}^*) = 0$, if the system remains at $\mathbf{s}^*$ for all $t \geq 0$ then $\mathbf{s}^*$ is an equilibrium. Equilibria can be stable or unstable, the stability of an equilibria can be probed by analyzing the response of the system near $\mathbf{s}^*$ to a small disturbance $\tilde{\mathbf{s}} = \mathbf{s} - \mathbf{s}^*$,

such that $|\mathbf{s}^* - \mathbf{s}| \ll 1$. Note that by definition $F(\mathbf{s}^*) = \mathbf{0}$, so $\mathbf{s}^*$ is treated as a constant, and $\dot{\tilde{\mathbf{s}}} = F(\mathbf{s}^* + \tilde{\mathbf{s}})$. Then, in time domain, we are interested in $\varphi^t(\tilde{\mathbf{s}})$, for some time $t > t_0 = 0$.

However, the trajectory based on f, g usually cannot be calculated when they are nonlinear (as is the case for 1.18). Then we use the linearization of the system near the equilibria, solving

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \equiv \begin{pmatrix} \dot{\tilde{v}} \\ \dot{\tilde{u}} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial f(\mathbf{s}^*)}{\partial v} & \frac{\partial f(\mathbf{s}^*)}{\partial u} \\ \frac{\partial g(\mathbf{s}^*)}{\partial v} & \frac{\partial g(\mathbf{s}^*)}{\partial u} \end{pmatrix}}_{\mathrm{J}} \begin{pmatrix} \tilde{v} \\ \tilde{u} \end{pmatrix} \tag{1.13}$$

note that in the linearization of the system quadratic and higher orders terms are omitted since they are evaluated on the small disturbance $\tilde{\mathbf{s}}$, and so are assumed to contribute extremely little to the solution.

To solve the system above we can use the eigenvalue method (if the eigenvectors are linearly independent) where solutions are in the form $\tilde{\mathbf{s}}(t) = e^{\lambda t}\boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the eigenvector. substituting into 1.13 we get

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \to \lambda e^{\lambda t}\boldsymbol{\xi} = \mathrm{J}e^{\lambda t}\boldsymbol{\xi} \to \mathrm{J}\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \tag{1.14}$$

The eigenvalues of J are found solving the characteristic equation $\det(\mathrm{J} - \lambda\mathrm{I}) = 0$, for 2×2 matrices, this leads to a quadratic polynomial that is solved as

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \tag{1.15}$$

where $\tau = \mathrm{Tr}(\mathrm{J})$, $\Delta = \det(\mathrm{J})$, note that each eigenvalue will have its corresponding eigenvector, so our example system has $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$.

Finally, the eigenvectors of J form the eigenbase of the system. That is, their linear combination gives the linear approximation of $\varphi^t(\tilde{\mathbf{s}})$:

$$\dot{\tilde{\mathbf{s}}} = c_1 e^{\lambda_1 t - t_0}\boldsymbol{\xi}_1 + c_2 e^{\lambda_2 t - t_0}\boldsymbol{\xi}_2 \tag{1.16}$$

where c_1, c_2 are determined by $\tilde{\mathbf{s}}$.

Clearly, the behavior near the equilibria will be a linear combination of exponential motion along $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$ with direction determined by the signs of λ_1, λ_2. Each equilibria of a system can then be categorized by its eigenvalues.

The main types of equilibria relevant to this work are

- Nodes: real eigenvalues with the same sign. If $\lambda_{1,2} < 0$ the node is stable: trajectories converge to it. If $\lambda_{1,2} > 0$ unstable: trajectories diverge from it.
- Saddles: real eigenvalues with opposite sign. Thus always unstable.
- Focuses: complex-conjugate eigenvalues (when $\tau^2 - 4\Delta < 0$), trajectories spiral either inward or outward depending on stability. Sign of $\Re(\lambda_{1,2}) \neq 0$ determines the stability, while $\Im(\lambda_{1,2})$ determines the frequency of the rotation.

Knowing type and stability of equilibria, joined with knowing F, provides information on all possible behaviors of a system, given its initial state.

1.1.3 From Earth to neurons

Historically the first model used to study SR was that of 1.2, more generally expressed as a stochastic differential equation [10]

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x\left(a - bx^2\right) + \mu\gamma\cos(\omega t) + \sigma\eta(t) \tag{1.17}$$

where η is gaussian white noise, and $x(a - bx^2)$ cubic term, that for $a > 0$ has 3 fixed points. When defining the pseudo-potential as in 1.4 it gives the quartic polynomial that is the simplest realization of the double potential well (here ignoring forcing and noise component)

$$\begin{aligned}\Phi &= -\int -bx^3 + ax \\ &= \frac{b}{4}x^4 - \frac{a}{2}x^2 + C\end{aligned}$$

Intuitively, a neuron is a system that displays both quiescent and spiking behavior: that is, 2 stable states, separated by a "barrier" commonly called the voltage threshold. Note that the term threshold is hardly accurate: different current values can all elicit spikes in neurons depending on stimulation duration, some neurons respond to certain stimulation frequencies at currents lower than those needed to elicit a spike with a pulse-like stimulation, and others fire in response to hyperpolarizing currents (known as a rebound spike) [11, Chapter 7].

Unsurprisingly, a neuron can be modeled in the same form of 1.17 to show SR [12]. Interestingly, the same holds in more accurate neuron models.

1.1.4 The FitzHugh-Nagumo model

The FitzHugh-Nagumo (FHN) model [13, 14] is a 2D reduction of the Hodgkin-Huxley (HH) biophysical model [15], its attractiveness stems from being able to recreate the spiking behavior of neurons without the 4 differential equations that regulate the HH neuron, allowing for easier analysis of the model using the theory of dynamical systems.

The FHN model is defined by a fast and a slow variable

$$\frac{\mathrm{d}v}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I \tag{1.18}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.19}$$

with v representing the (fast) membrane voltage and w the (slow, with $\varepsilon \ll 1$) feedback recovery mechanisms of the membrane (e.g., opening of K^+ channels). The main difference of this model with 1.2 is the second equation for w that allows for an additional behavior that cannot be seen in the climate model. However, a cubic term for $\dot{v}$ is present and one may wonder whether the resulting double well potential is a requirement for SR. It will be shown that it is not, and that it would describe an unrealistic regime for neural function under SR.

To see how the 2 models are related, both can be analyzed with the methods of 1.1.2. In the case of $\varepsilon \to 0$, we can treat w as a constant, then it is clear that the FHN model's phase portrait is topologically equivalent to 1.17 (see fig) for the right $\mathcal{P}$ (see B).

However, the FHN 1d reduction does not have a spiking mechanism, and the resulting neuron model would only show 2 resting states at different membrane voltages values v_1, v_3, exactly as in 1.2. Thinking about biological neurons, we would be describing a neuron with a single ionic current with a depolarizing effect, so voltage would only grow, with no recovery feedback. Clearly the 1d model is insufficient. Inspecting the full FHN we note that it is the slow w, entering as a subtractive term in v, that causes additional repolarization of the membrane, mapping to our biological understanding of the spiking mechanism.

Then the nonlinearity we are interested in is whichever is born from the interaction of the 2 opposing forces.

1.1.5 Neuron types and inputs

Different $\mathcal{P}$ construct different fields F, resulting in different equilibria, both number and type, and bifurcations; all leading to different behaviors. Thus the FHN can model different types of neurons, the main types are known as "integrators" (type-1) and "resonators" (type-2). The 2 types have different computational properties, but both can rightly be used for primary and secondary sensory neurons (which are of interest in this chapter), depending on the type of stimulus they are supposed to receive and process.

Stimuli can be modeled as instantaneous pulse-like or smooth ramp-up/down inputs. It is useful to clarify what is intended for instantaneous input (or shift in v), as it is also the formalization of a spike event in more simplified models, that will be used later. The common mathematical object used is Dirac's δ

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \tag{1.20}$$

such that

$$\int_{-\infty}^{\infty} \delta(x)dx = 1 \tag{1.21}$$

Then, defining the stimulation time as t_0, 1.18 becomes

$$\frac{\mathrm{d}v(t)}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I(t) \tag{1.22}$$

$$\frac{\mathrm{d}w(t)}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.23}$$

where $I(t) = Q\delta(t - t_0)$, and Q is the voltage jump (more correctly Q/C, but the membrane's capacitance is assumed $C = 1$). Essentially $I(t)$ at t_0 shifts the whole system state vector to a new position in the state space $\tilde{\mathbf{s}} = (v + Q,\, w + 0)$, acting as a macro-perturbation from equilibrium.

It is to note that the δ-like input is one limited way to model the stimulation that a neuron may be subjected to. Often a neuron may receive multiple inputs from pre-synaptic elements that can vary in intensity, depending on many biological details. Thus, another way to model stimulation is to allow $I(t)$ to vary continuously with time, mimicking the smoothing summation of incoming inputs at different timings.

From the perspective of dynamical systems, the effect is not of a perturbation on $\mathbf{s}$. Rather, when the changes in I are slow enough, the geometry of the phase portrait itself changes, as w changes accordingly. Effectively I is to be thought of as another parameter in $\mathcal{P}$ that we choose to vary.

As seen in 1.1.2, this is the setting for a codimension-1 bifurcation. Then, unlike δ inputs, smooth ones result in a change in the topology of the phase portrait of the system, changing its overall behavior.

In 1.1.1 we saw that SR happens for an input $\mu(t)$ that was unable by itself to lift the system over the pseudopotential "wall" (intuitively no better tracking of $\mu(t)$ could be made otherwise). It is clear that for neurons the "wall" is the difference between the spiking and non-spiking state. Therefore, it is now necessary to clarify the meaning of threshold.

1.1.6 Neural nonlinearity, or threshold phenomena

The usual working definition of a threshold is a membrane voltage value v_ϑ that when reached is followed by a large excursion in v, while no excursion is present for $v < v_\vartheta$. The discontinuity in the trajectories of v near v_ϑ is our wall to be crossed. However, looking at

the vector fields depicted in the phase portraits, it is unclear whether there really is such a threshold value.

Indeed, consider the general system $\dot{\mathbf{x}} = f(\mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^n$, and the flow $\varphi^t(\mathbf{x}_0)$ defined on it. It can be proven (see [16, Chapter 2]) that if the partial derivatives of the system are continuous in an open set $X \subset \mathbb{R}^n$, then it is locally Lipschitz on X, that is, for every point $\mathbf{x}_0 \in X$ there exist a set $N_\varepsilon(\mathbf{x}_0) \subset X$

$$N_\varepsilon(\mathbf{x}_0) = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{x}_0| < \varepsilon\} \tag{1.24}$$

such that for $\mathbf{x}_1$, $\mathbf{x}_2 \in N_\varepsilon(\mathbf{x}_0)$ and constant L

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq L\,|\mathbf{x}_1 - \mathbf{x}_2|\,. \tag{1.25}$$

Then if $f(\mathbf{x})$ is continuously differentiable on a closed invariant set, for arbitrarly close starting points the trajectory divergence is bounded by Gronwall's inequality to

$$|\varphi^t(\mathbf{x}_1) - \varphi^t(\mathbf{x}_2)| \leq e^{Lt}\,|\mathbf{x}_1 - \mathbf{x}_2| \tag{1.26}$$

Now consider the FHN in its simplest configuration: having a unique equilibrium that is a stable node (see fig[FHN unique eq]), and t^* as a fixed time at which we expect the action potential peak to be, and $Q \subset \mathbb{R}$ as the open set of input values. The v-nullcline divides the state space in regions of $\dot{v} > 0$ and $\dot{v} < 0$: under it voltage grows quickly before coming back to the stable node, while above it it decreases. Then a spike would be a trajectory that travels to the right of the unstable branch of the v-nullcline and crosses it after the right knee, while a subthreshold response would cross the v-nullcline almost immediately, with no visible spike.

To simplify our analysis, we consider our system starting at the stimulation time t_0, $\mathbf{s}_0 = (v_0, w_0)$ the stable node coordinates, and $\gamma_0(q)$ giving the initial state as a continuous function of the input $q \in Q$ as

$$\gamma_0 : Q \to \mathbb{R}^2,\ q \mapsto (v_0 + q, w_0) \tag{1.27}$$

Having a single asymptotically stable equilibrium and polynomial derivative the system is bounded in a trapping region K, hence satisfaying the Lipschitz condition. Therefore, if we wanted to identify a $q_\vartheta \in Q$ such that a discontinuity (almost instantaneous divergence for $t^* > t_0$) exists between $\varphi^{t^*}(\gamma_0(q_\vartheta))$ and $\varphi^{t^*}(\gamma_0(q_1))$ for any arbitrarly close q_ϑ, $q_1 \in Q$ we would be out of luck.

That is, across Q we will encounter trajectories that travel along the unstable branch up to the right knee before going to the $\dot{v} < 0$ region (also called canards): those are intermediate amplitude responses.

Appendix A

Stability of Equilibria

In Benzi et al. [5]

Appendix B

Phase portraits of climate and FHN models

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